Human Imitator

Hi, I manage a local bike repair shop. We sell bicycle inner tubes and repair patches daily, but we face unpredictable demand—sometimes we have too many stockpiled, other times customers can't get what they need quickly enough. It's challenging to balance ordering enough without tying up capital in unsold items while also ensuring timely service. Can you assist us in creating a better inventory strategy?

LLM Consultant

Of course, I'd be happy to help you develop a more effective inventory strategy. Let's start by gathering some key information to better understand your current situation and needs.

1. **Demand Information**: Could you tell me the average daily demand for the bicycle inner tubes and repair patches? Also, do you have any information on the daily standard deviation of the demand for these items? This will help us understand the variability and unpredictability you're experiencing.

2. **Lead Time**: How long does it typically take for your orders to arrive once you've placed them with your supplier?

3. **Current Inventory Levels**: What are your current inventory levels for both inner tubes and repair patches? How often do you currently place orders?

4. **Service Level Requirement**: What level of service are you aiming for? For example, how important is it for you to ensure that you never run out of stock? Are you aiming for 95% service level or another target?

5. **Order Costs and Holding Costs**: Do you have any information on the costs associated with placing an order and the costs of holding inventory? This includes any storage costs or capital costs.

Once I have this information, I can help you devise a strategy to better align your inventory with your demand.

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Average daily demand is around 8 units, with a standard deviation of 2 units due to the Poisson distribution. Lead time is approximately 10 days from when we place an order with our supplier.
We usually keep about 10 units in stock at any given time and review inventory every week to decide if we need to reorder based on expected demand.
Our primary goal is to maintain a high service level—ideally above 90%. Customers expect quick turnaround times since we're a repair shop.
Holding cost is $0.5 per unit per day, which covers storage space and potential spoilage (though there's no expiration date). The main penalty comes from delayed repairs; each backlogged patch costs roughly $2.83 per day per item held over.
There's a fixed setup cost of $3 per order regardless of quantity ordered. There isn't much flexibility beyond those constraints—the maximum inventory allowed is 80 units, and we can order up to 25 units per batch.

LLM Consultant

Thank you for providing detailed information. Based on what you've shared, we can develop an inventory strategy that helps you maintain a high service level while minimizing costs. Here's a structured approach using an (s, S) inventory policy:

1. **Reorder Point (s)**: This is the inventory level at which you should place a new order. To calculate this, we need to consider both the average demand during the lead time and the safety stock needed to maintain your desired service level.

   - **Average Demand During Lead Time**: \(\text{Average daily demand} \times \text{Lead time} = 8 \times 10 = 80 \) units.
   - **Safety Stock**: To achieve a 90% service level, we can consider the standard deviation of demand over the lead time. Given the daily standard deviation is 2 units, the standard deviation over 10 days is \(\sqrt{10} \times 2 = 6.32 \) units.

   Using a safety factor (z-score) for a 90% service level, which is approximately 1.28:
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   \text{Safety stock} = 1.28 \times 6.32 \approx 8.1 \text{ units}
   \]

   - **Reorder Point (s)**: \(80 + 8.1 \approx 88.1 \), rounded up to 89 units to be conservative.

2. **Order-Up-To Level (S)**: This is the level to which you will restock your inventory after placing an order.
   - Given your maximum inventory level is 80 units, and you can order up to 25 units at a time, let's set the order-up-to level based on your constraints. Since your inventory cannot exceed 80 units, you'll order up to 80 units.

3. **Ordering Strategy**:
   - Monitor your inventory level daily.
   - When your total inventory (on-hand plus on-order) drops to or below 89 units, place an order to bring the inventory up to 80 units, respecting the maximum order quantity of 25 units.

4. **Review and Adjustments**:
   - Regularly review your service level and inventory levels. If you find that stockouts are more frequent than expected, consider adjusting the safety stock or reviewing the variability in demand.

In summary, your strategy is:
- **Reorder when inventory falls to 89 units or lower**.
- **Order enough to reach up to 80 units**, considering a maximum order batch of 25 units.

This approach should help you maintain a high service level of above 90%, balancing customer satisfaction and cost management. Let me know if you need more clarification or further adjustments!